

Do LLM-Written Dictionaries Resemble Human Ones? Mutual Definitions with and without Instruction Tuning

Claims sheet for Kumoi, draft of 2026-10-02, version 2. Working title (D028). A dated snapshot of claims 1-3 in research/PROJECT_CONTEXT.md, which stays the record.

Question and short answer

Question. Do LLM-written dictionaries close like human ones, and is that structure created by instruction tuning rather than by scale?

Short answer. Every instruct model defines words by each other well above chance, of the same order as WordNet (R 8.8–42.5 against 15.7), though mostly with different pairs. The hypothesis that instruction tuning creates this is not supported by these tests: given three example definitions, Gemma3 base models at 4B–27B show no difference from their instruct counterparts' R, and at 1B they reach about half of it.

Setting. 17 instruct models (Gemma3-270M instruct excluded: no cycles, D011), 5 Gemma3 base models and WordNet; 3,000-word list (2,750 lemmas); an edge from each word used in a definition to the word defined; null = degree-preserving rewiring (32× edges swaps, 100 replicates); R = observed mutual pairs / null mean. Default filter and first sentence unless stated.

Claims

- Kernel ratio is at chance after density correction; circulation rate is below chance for the instruct models and WordNet.**
 - Kernel z, instruct: median +0.16 unfiltered, +0.31 default filter, +0.10 strict (WordNet −0.08; two or three models below −2, one above +2). Circulation z, instruct: median −1.34 / −1.51 / −1.37, 15 of 17 below 0 and 6 below −2 in each variant (WordNet −1.89). The five base graphs are too sparse for either measure to be stable (circulation z median −1.81 unfiltered, +1.90 default, +2.20 strict). Core and MinSet were not tested.
- One structure remains: reciprocity excess R.**
 - R, default filter: instruct 8.8–42.5 (median 19.3), zero-shot base 2.1–8.4 (4.9), WordNet 15.7. By WordNet relation the mutual pairs are 38% synonyms, 22% hypernym pairs, 9% sister terms, 2% antonyms, 1% derived forms and 29% with no direct link; several hypernym and unlinked pairs are near-synonyms in ordinary use (flavor–taste, location–place).
 - LLMs and WordNet agree in the size of the excess, not in which words form it: each instruct model shares 1–4 of WordNet's 27 pairs (Jaccard 0.010–0.055); two instruct models share more (median 0.14).
- "Instruction tuning, not scale, creates it": not supported as stated.**
 - Zero-shot, the matched Gemma3 pairs still separate: instruct R / base R 2.2–13.9 across five variants of the analysis (default 7.1 / 3.3 / 12.2 / 5.2 at 1B / 4B / 12B / 27B). A stricter filter, removing six frame words and using the whole stored text do not close the gap; its cause is not identified.
 - With three examples (E3), instruct R / base R is 1.91 / 1.19 / 0.83 / 0.89 (exact 95% ranges [1.13, 3.24], [0.82, 1.71], [0.61, 1.12], [0.64, 1.25]). At 4B–27B no difference can be shown (the ranges still allow up to about 1.6–1.7×); at 1B the base model reaches about half by the exact method, but the conservative range [0.92, 3.98] includes 1. Gemma3-270M has no instruct comparison; its few-shot base R is 4.6 [2.3, 8.3], below every instruct model.
 - In a blind hand check of 100 E3 outputs (25 words per model, 1B–27B) all were definitions in form (95% interval 96–100%); some are circular or wrong, so this certifies form, not accuracy. Their mutual pairs overlap the instruct models' pairs: Jaccard with the matched instruct model 0.15 / 0.17 / 0.22 / 0.20, inside the range between two Gemma3 instruct models (0.12–0.28, median 0.21).
 - Scale (descriptive, not a test): among the 17 instruct models R shows no overall relation with size (Spearman +0.18); it falls within Gemma3 (42.5 at 1B, 25.4 at 27B), has no trend within Qwen2.5 and rises within Qwen3 (8.8 at 0.6B, 29.3 at 8B). Few-shot base R rises from 4.6 (270M) to 22–31 (1B–27B), and the instruct advantage under few-shot shrinks from 1.91 to 0.89. The three E3 examples (book, walk, large) define by category and features, not by synonym, so they do not teach the property measured.

Limits we state

- One base family, one prompt, one word sample, one generation run per model (temperature 0.7; run-to-run variation unmeasured). No instruct model was run with the three examples, so the E3 comparison is few-shot base against zero-shot instruct.
- Filtered base graphs hold 3–14 mutual pairs; the count-based ranges treat pairs as independent although pairs share words (fifteen–sixteen, fifteen–fourteen), so they may be too narrow. Weakest zero-shot pair: 4B, 3.3 [1.7, 7.1].
- The filter catches 81.5% of base-model non-definitions (77.7% over all models), precision 97.5% on held-out words. One labeler did all hand labels (self-agreement kappa 0.90; D022: the file times cannot confirm the one-day gap between labelings, the researcher states that about a day passed, so the agreement is called a possible upper bound).
- Both prompts ask for common words and no repeated headword, which cannot explain why examples close the gap but bears on the comparison with WordNet (untested). WordNet glosses are not a dictionary written for people, and the graphs cover an open 2,750-word sample, which may be why the kernel cannot tell WordNet from a random graph.
- Two of the three example headwords (book, large) and ten words of the example definitions are in the vocabulary; edges from them are 5.1–5.5% of E3 edges at 4B–27B, as in the instruct graphs, and one E3 pair involves a headword (great–large, 270M). E4 and E5 are not hand-checked; E8 was not run.

Decisions

Confirmed (2026-10-02): D003, D005, D007 and D020, D012 and D016, D013 (no Qwen base replication), D015. **Open:** framing and wording of the conclusion (D026: "not supported by these tests", not "proven false"). **For information:** D018 label conventions, D019 extraction rule, D021 converged re-run of E3–E5, D022 self-consistency re-label, D023 filter, D024 strict bracket, D025 frame words, D027 E3 hand check, D028 title.

Sources: research/PROJECT_CONTEXT.md, EXPERIMENT_REGISTRY.md, DECISION_LOG.md, RESEARCH_AUDIT.md; audit_scripts/outputs 17 (sections 6 and 14), 18, 19 (section 6), 20, 22, 23.